

Assignment # 1**Applied Physics**

Max. Marks: 60

Section: BCS 1B

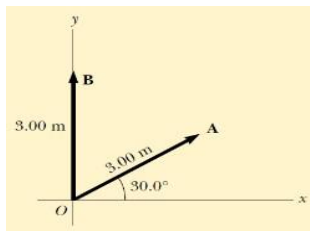
Submission Date: 15th September, 2022 (Thursday)**Instructions:**

1. First page of the assignment should be the page given below
2. Assignment should be hand-written. Typed assignment will not be accepted.
3. Use A4 Paper for the assignment. Do not use any other sheet
4. Please write the answers in the space given below.
5. Make diagrams wherever possible.

ROLL NO.: - -----NAME: -----

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Question #1: Each of displacement vectors A and B shown in the Fig. has a magnitude of 3.00m. Find graphically (a) $A+B$, (b) $A-B$, (c) $B-A$, (d) $A-2B$. Report all angles counterclockwise from the positive x-axis. (10 Marks)

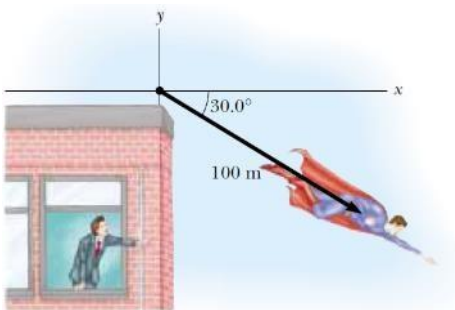
**Answer:**

Question 2: Discuss uniform circular motion to calculate centripetal acceleration, time period, scalar components of velocity vector, acceleration vectors and the angle between x and y components of acceleration. Draw respective diagrams showing displacement vectors, velocity vectors and acceleration vectors. (5 Marks)

Question #3: You are working at a tropical resort, and are setting up a treasure hunt activity for the guests. You've been given a map and told to follow its directions in order to bury a "treasure" at a specific location. You don't want to waste time walking around the island, because you want to finish early and go surfing. The directions are to walk 3.00 km headed 60.0° north of due east and then 4.00 km headed 40.0° north of due west. Where should you head and how far must you walk to get the job done quickly? Find your answer (a) graphically and (b) using components. (10 Marks)

Question #4: A boy stands at the edge of a cliff and launches a rock upward at an angle of 45° . The rock comes back down to the elevation where it was released 2.25 s later, then continues until it is seen to splash into the lake below 4.00 s after release. How far below the point of release is the lake surface? (10 Marks)

Question #5: Find the horizontal and vertical components of the 100-m displacement of a superhero who flies from the top of a tall building following the path shown in Fig. (5 Marks)



Question 6: Solve the Sample Problems: 3.03 and 4.05 from the Textbook (5+5=10 marks)

Question #7: A commuter airplane takes the route shown in Fig. First, it flies from the origin of the coordinate system shown to city A, located 175 km in a direction 30.0° north of east. Next, it flies 153 km 20.0° west of north to city B. Finally, it flies 195 km due west to city C. Find the location of city C relative to the origin.

(10 Marks)

